

The General Modeling Problem

Machine Learning as Modeling

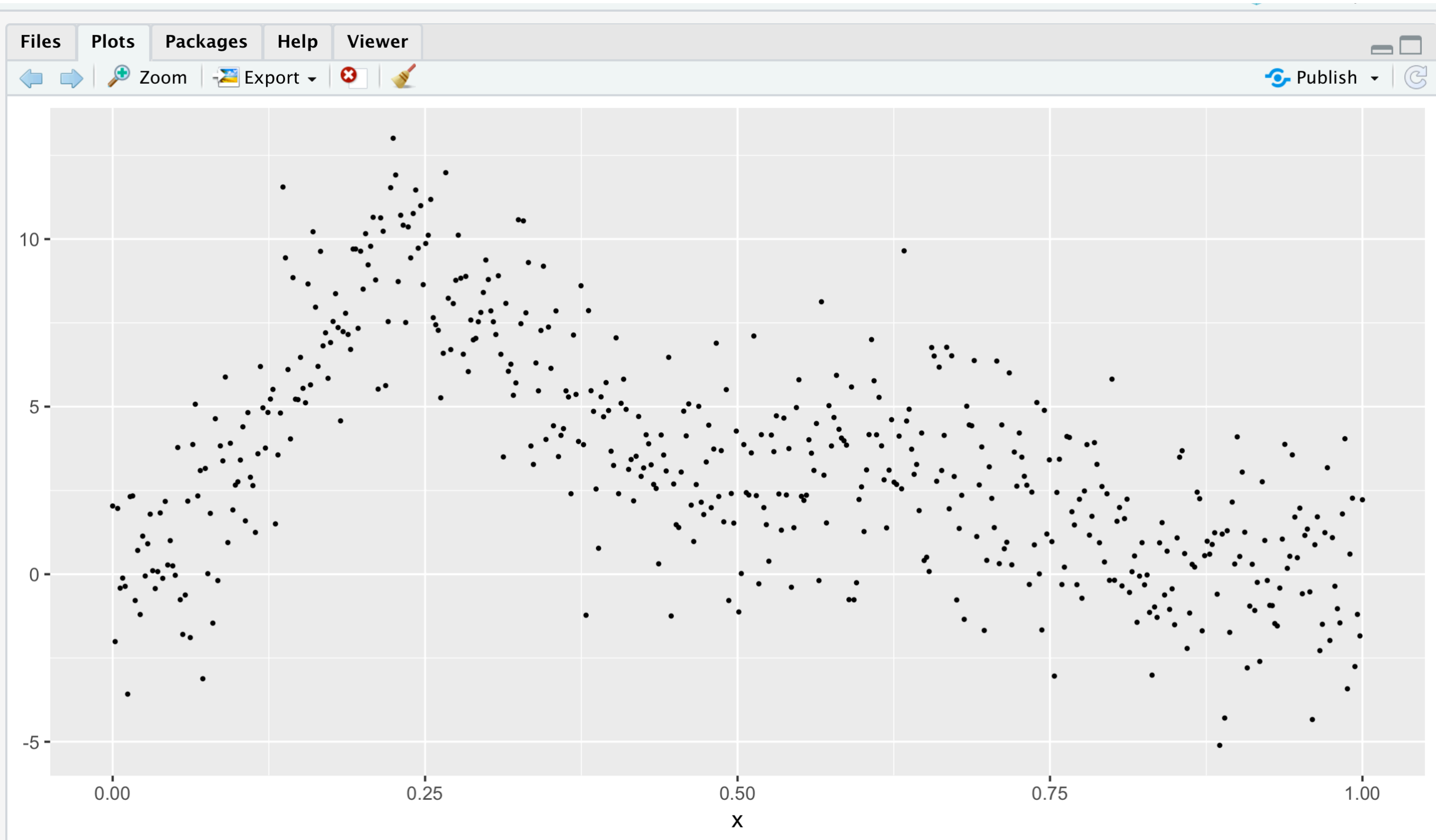
True (Unknown) Model: $y = f(\vec{x}) + \epsilon$

Approximated Model: $\hat{y} = \hat{f}(\vec{x})$

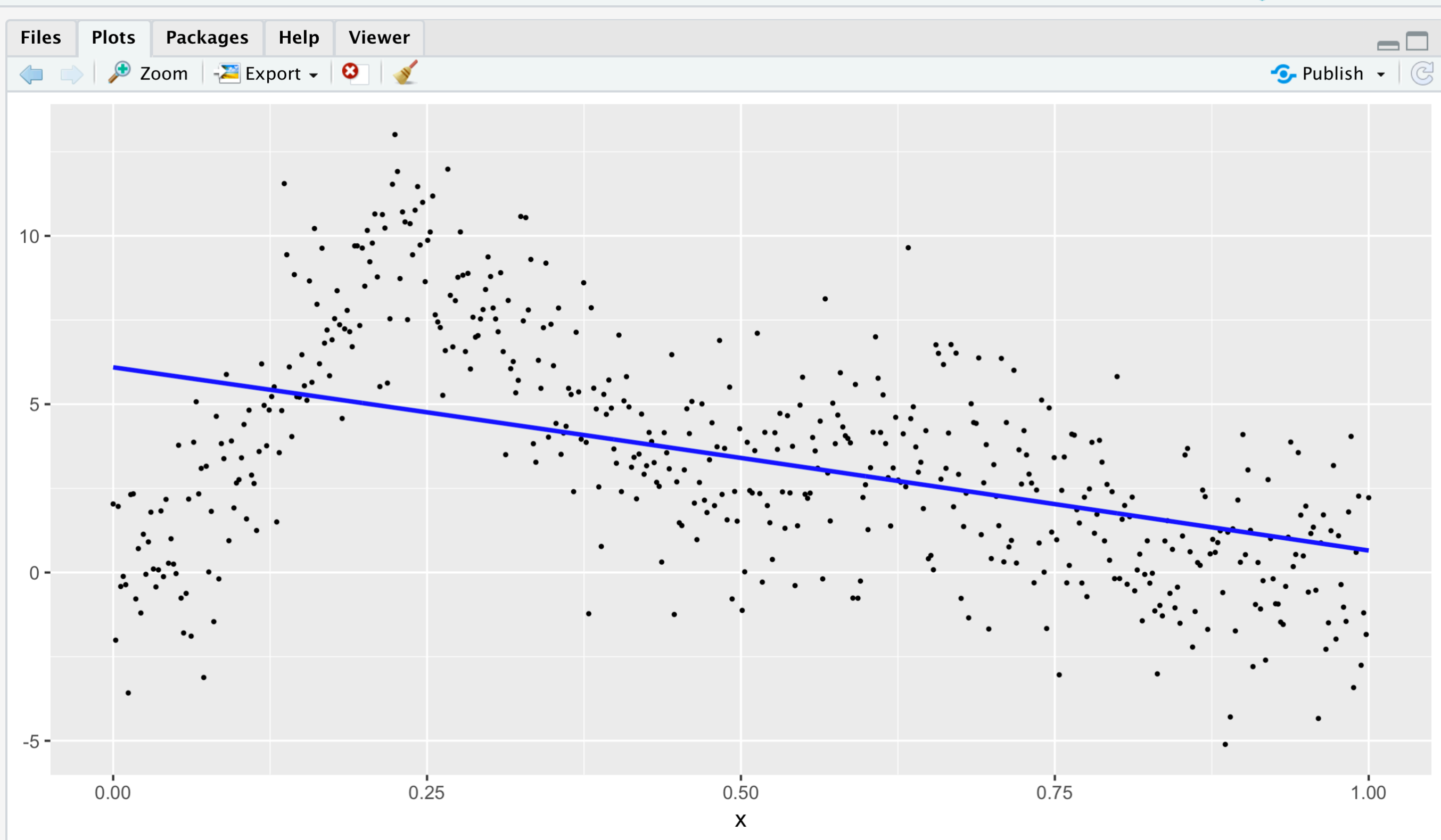
Now to the blackboard for
Chalk Talk #1...



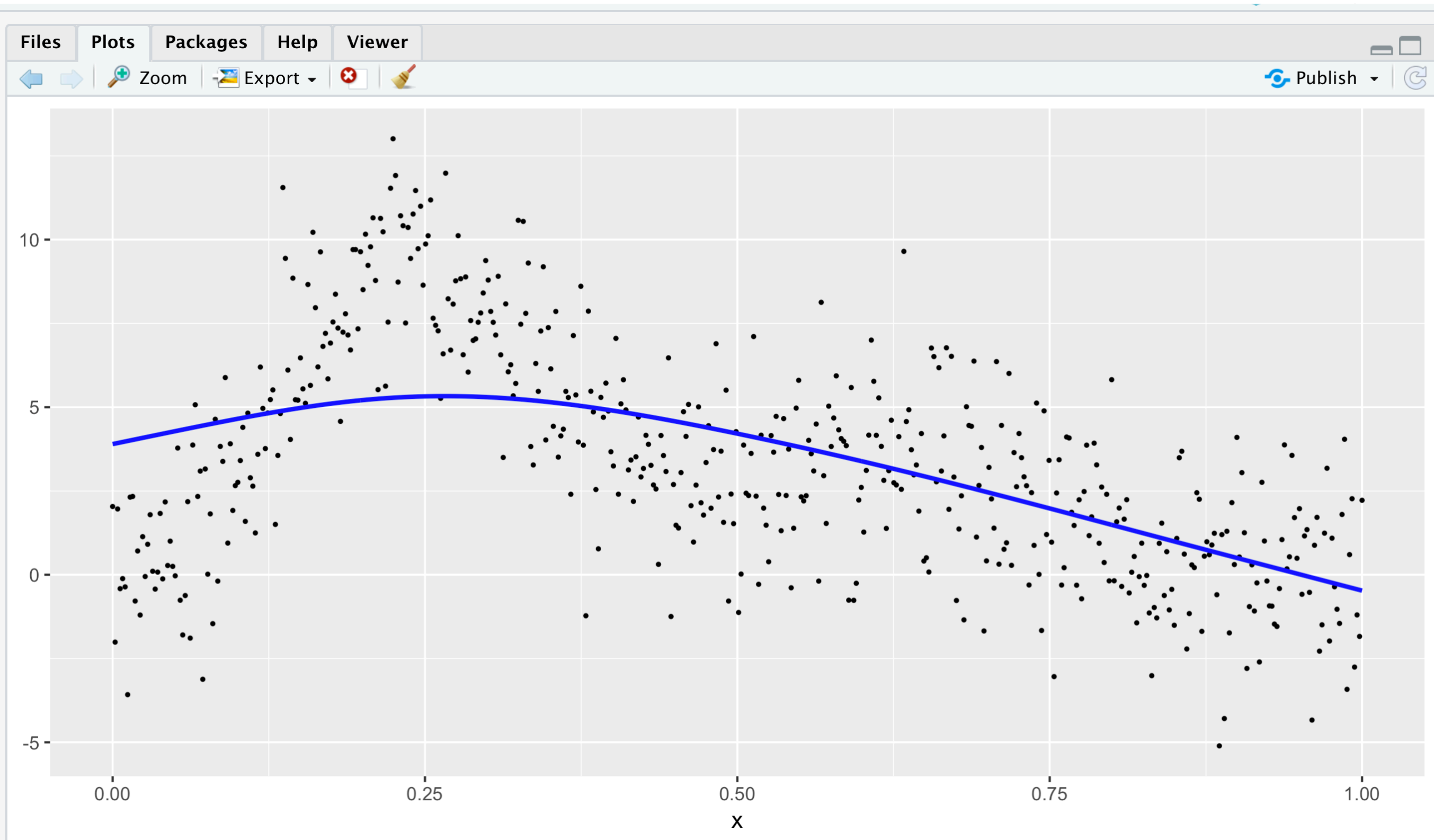
Given Data (x, y) from “unknown” $f(x)$



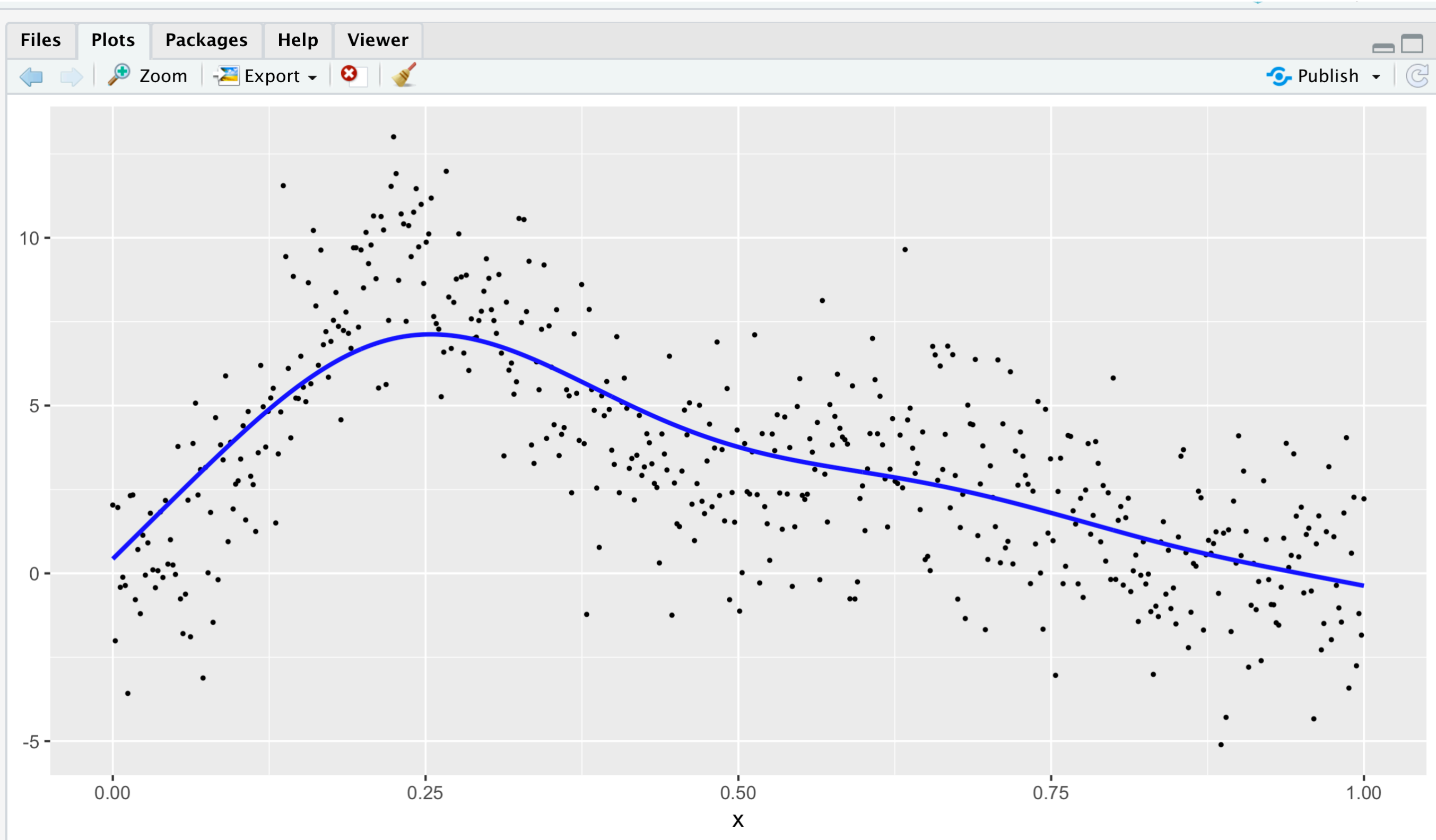
Approximate (i.e. “fit”) a Model $\hat{f}(x)$



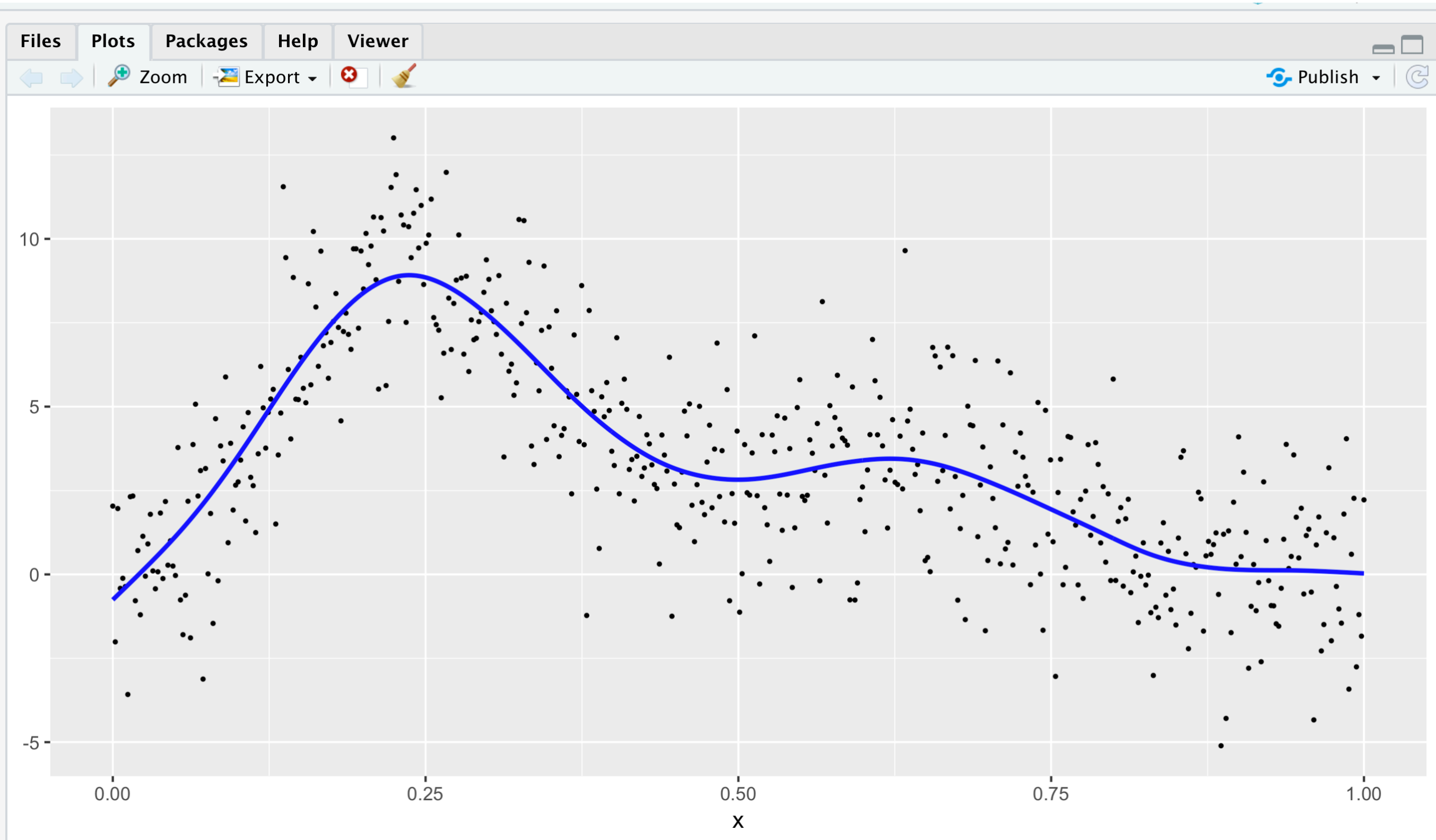
How about this $\hat{y} = \hat{f}(x)$?



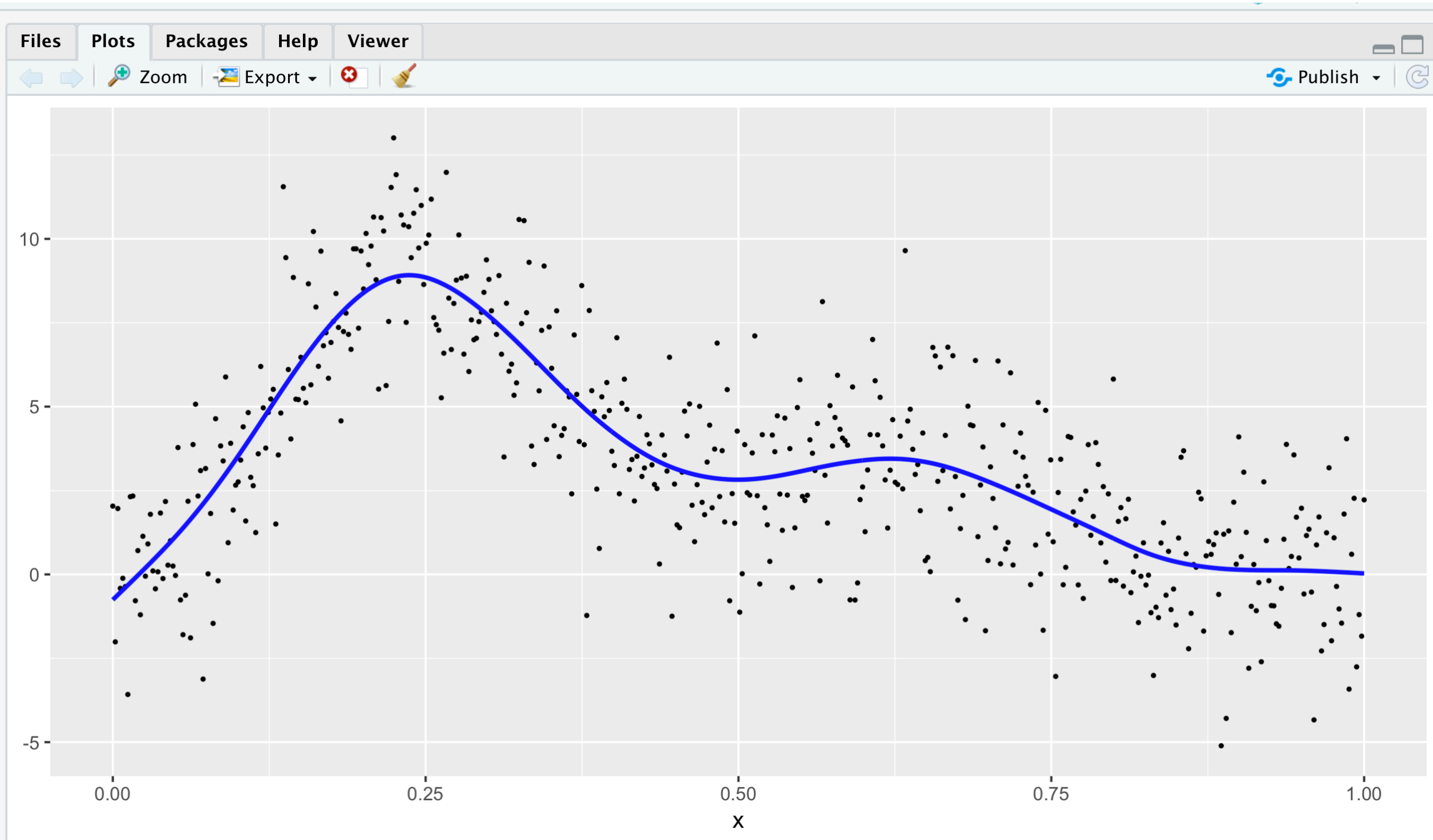
How about this $\hat{f}(x)$?



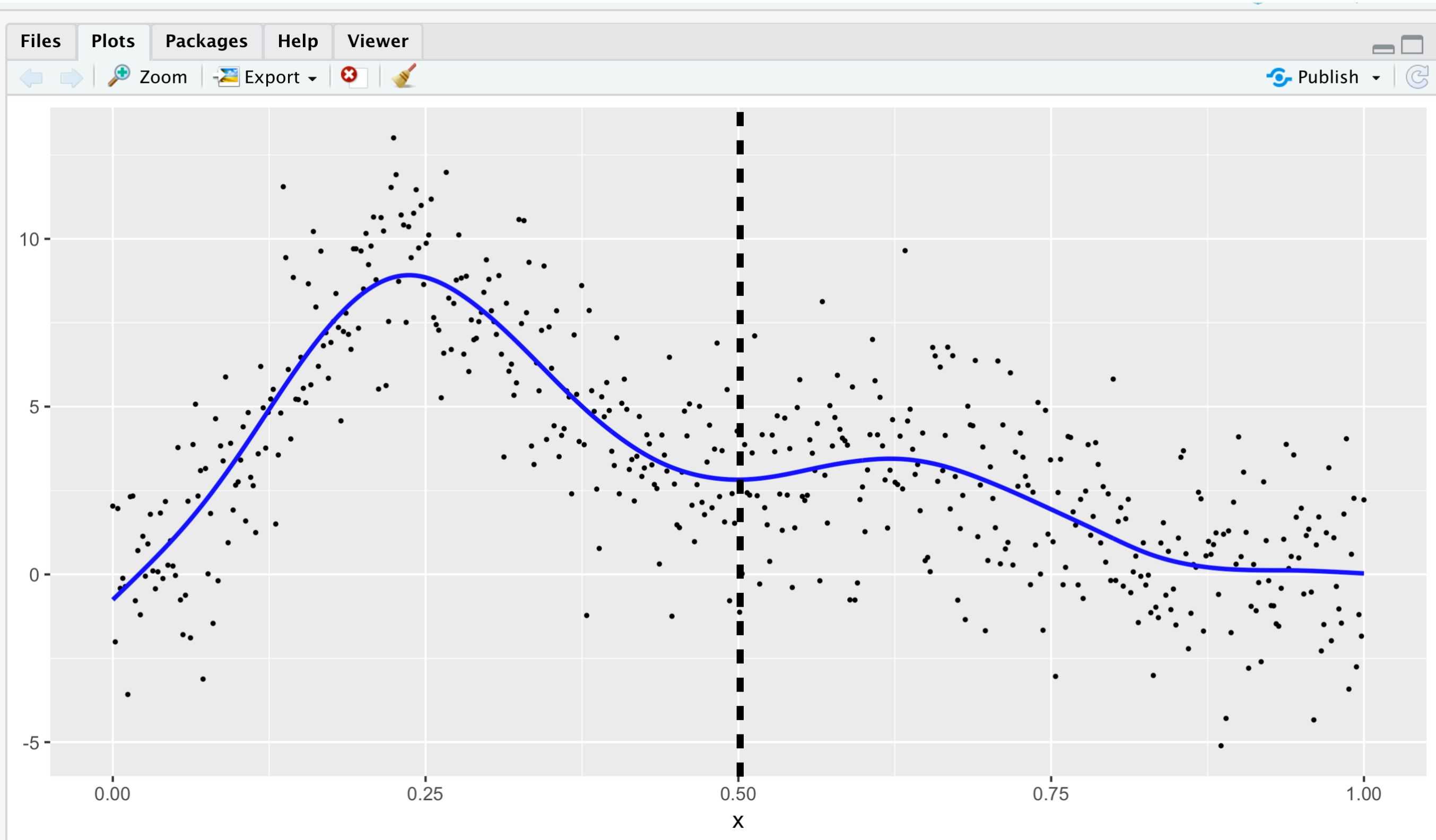
How about this $\hat{f}(x)$?



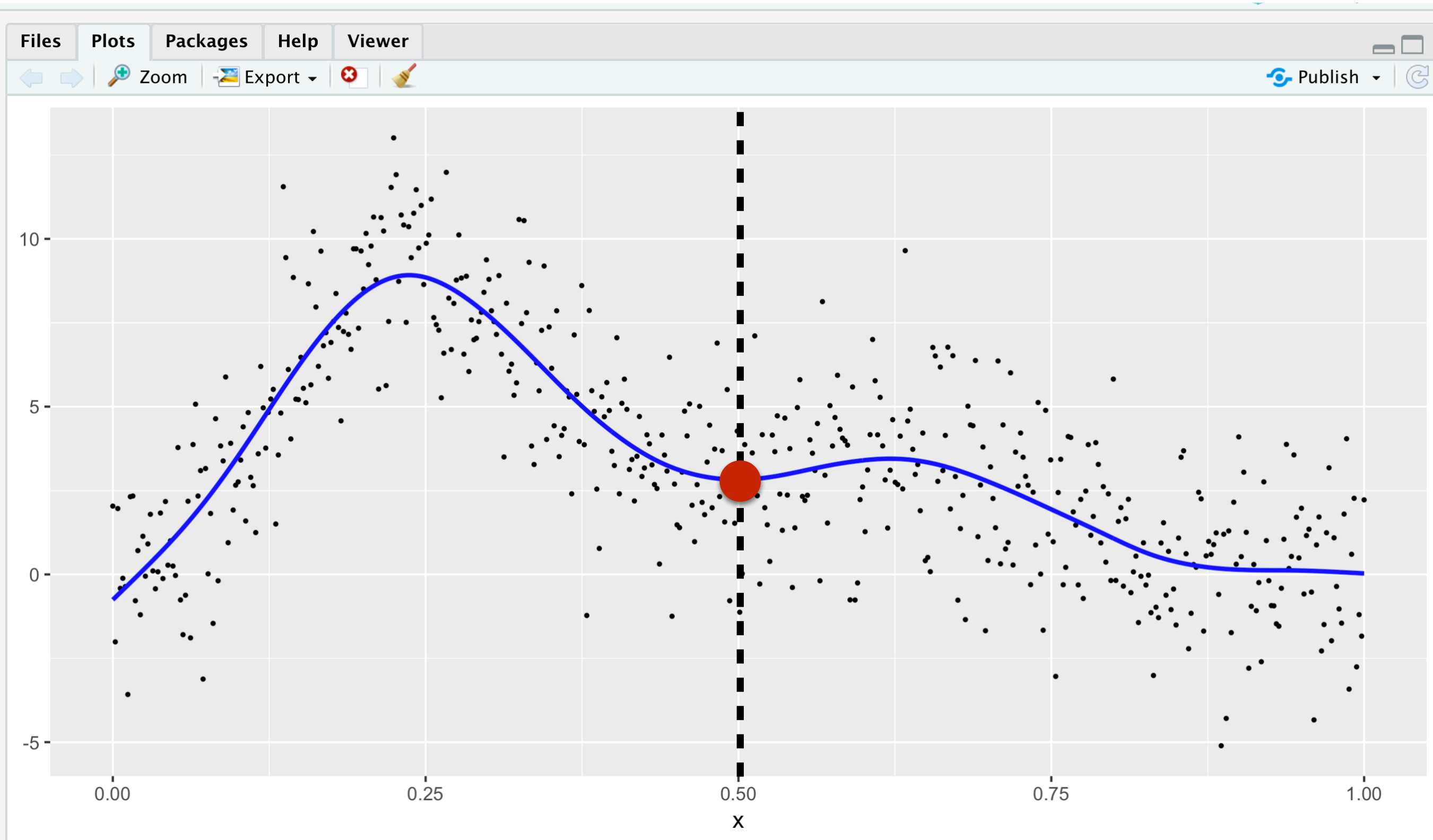
What does this $\hat{f}(x)$ predict for $x = 0.5$?



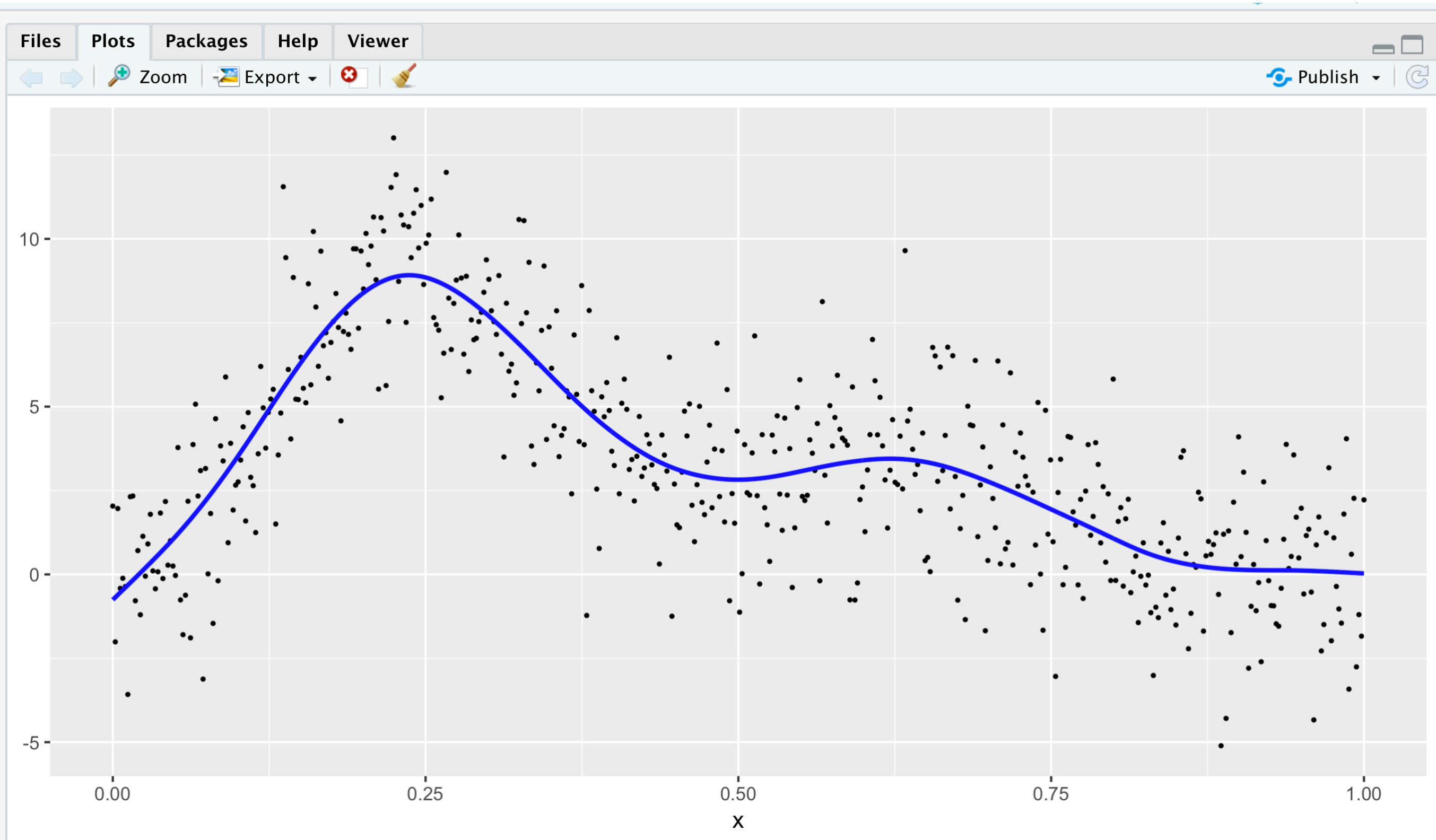
What does this $\hat{f}(x)$ predict for $x = 0.5$?



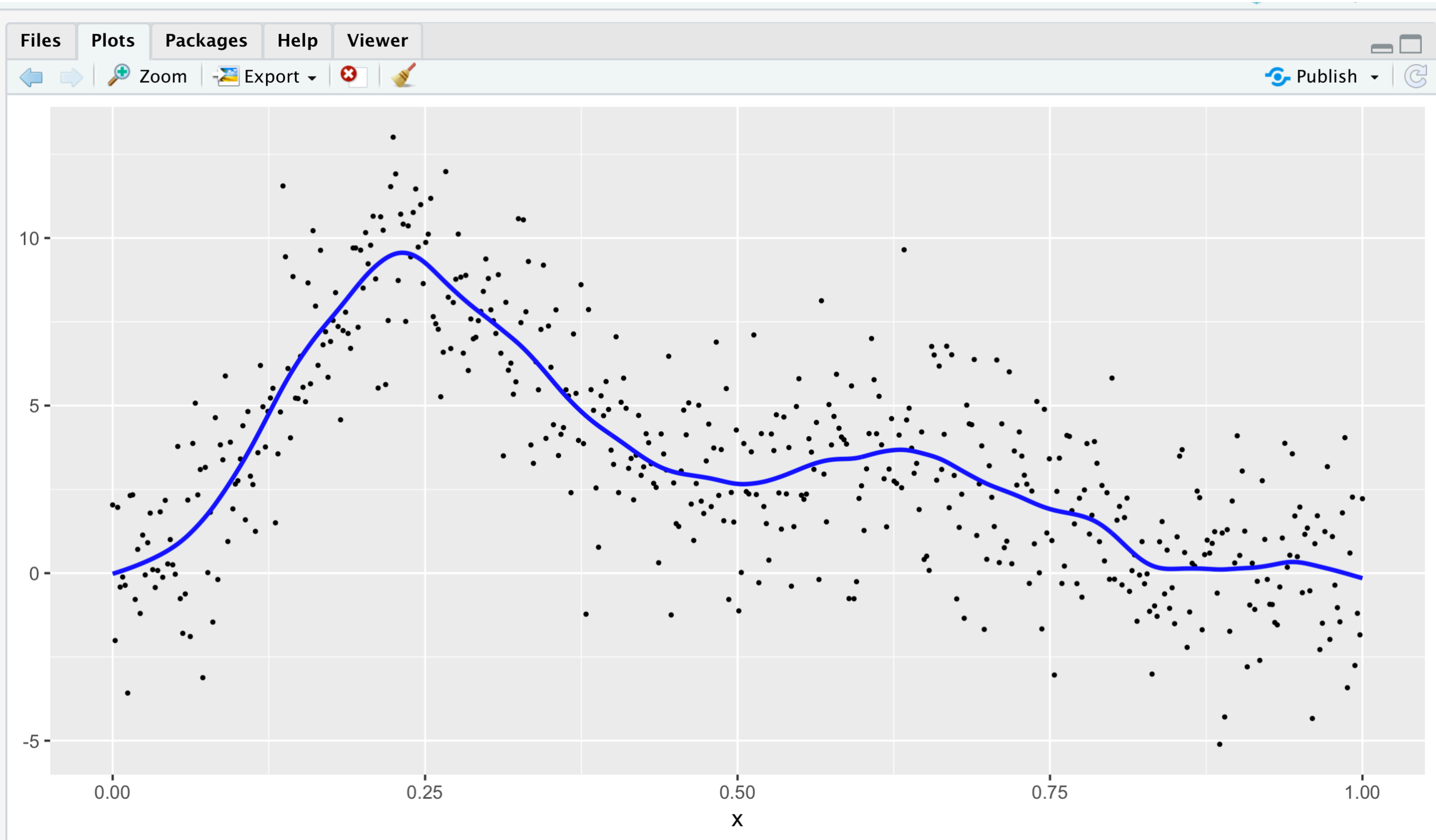
What does this $\hat{f}(x)$ predict for $x = 0.5$?



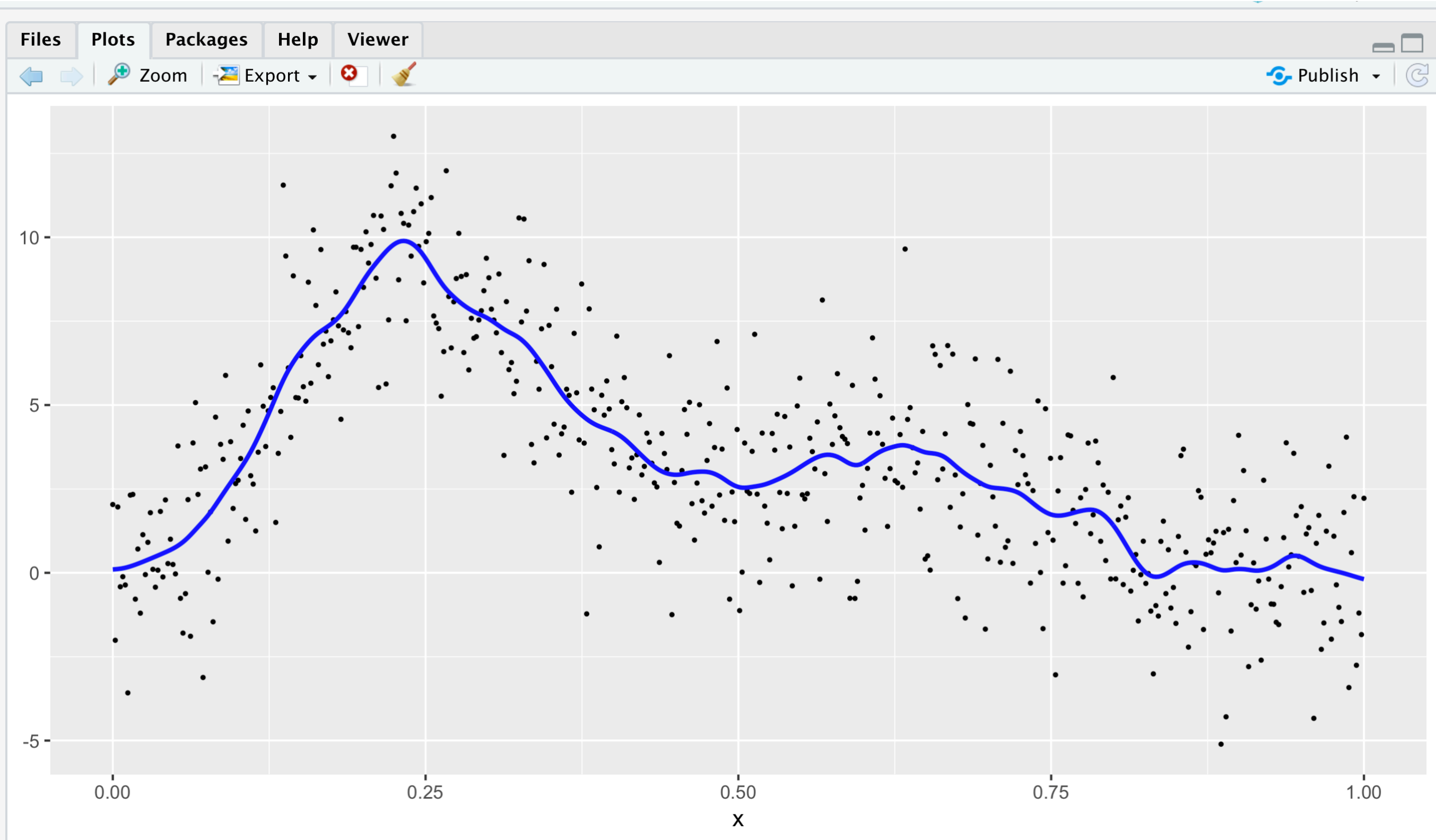
Ok, great. But instead of this $\hat{f}(x)$...



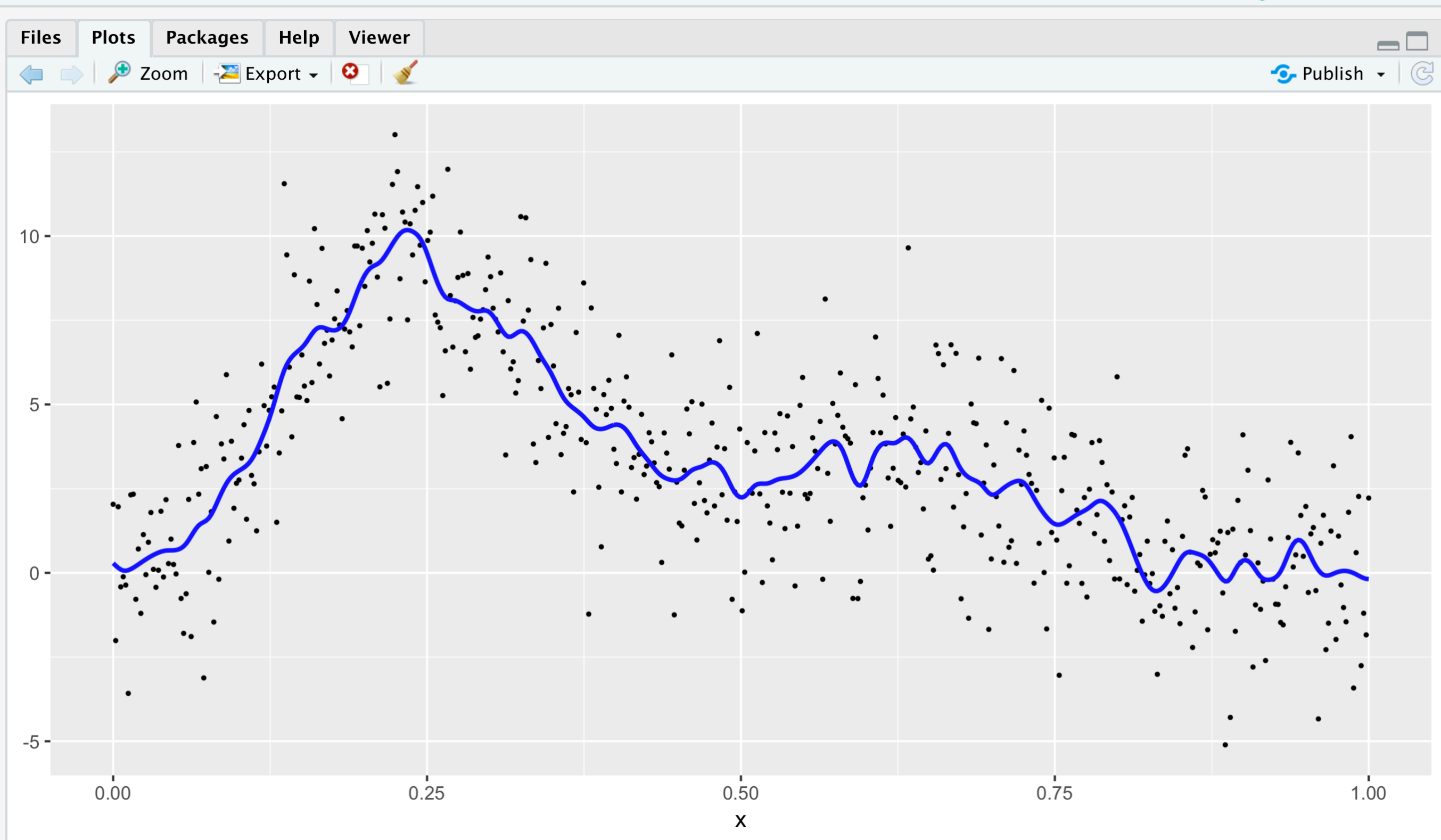
How about this $\hat{f}(x)$?



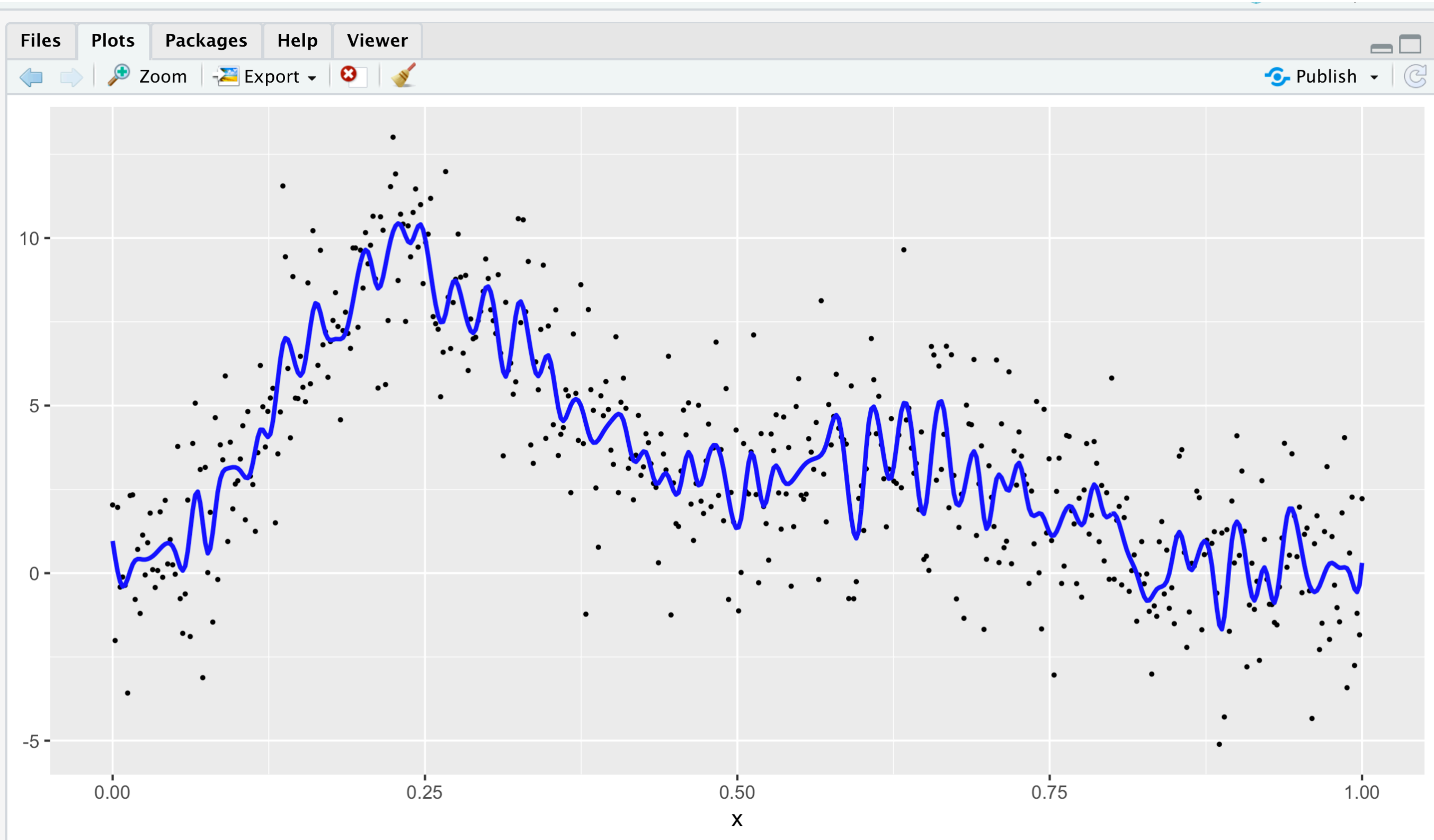
How about this $\hat{f}(x)$?



How about this $\hat{f}(x)$?



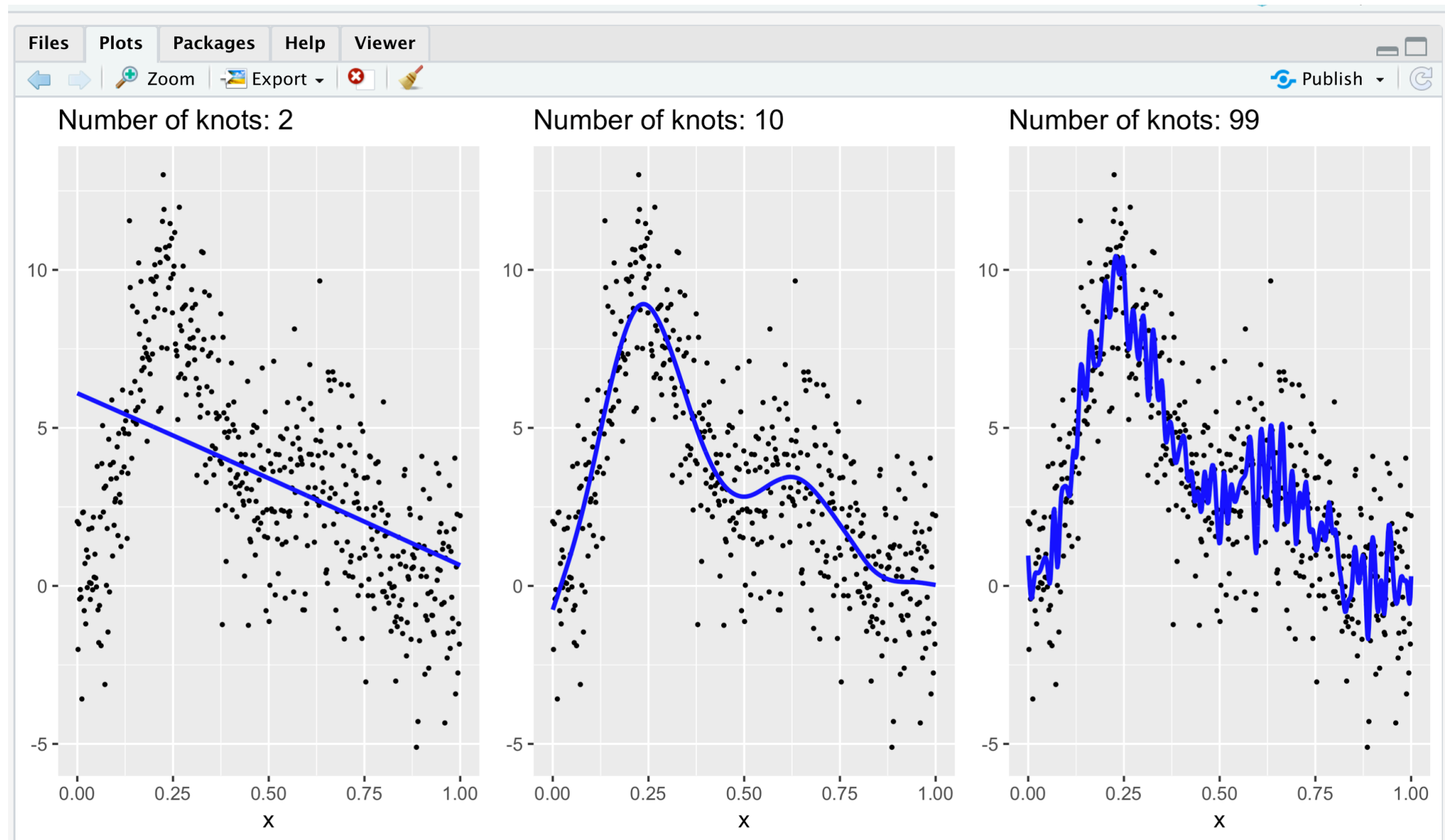
How about this $\hat{f}(x)$?



Model Fitting Method: (Cubic) Splines

- Splines fit the blue curve $\hat{f}(x)$ that **minimizes** the (squared) vertical distances between all:
 - predicted points $\hat{y} = \hat{f}(x)$ and
 - observed points y
- Amount of “wobble” is the **complexity of the model**
- Occam’s Razor

Three Different $\hat{f}(x)$



Underfit!

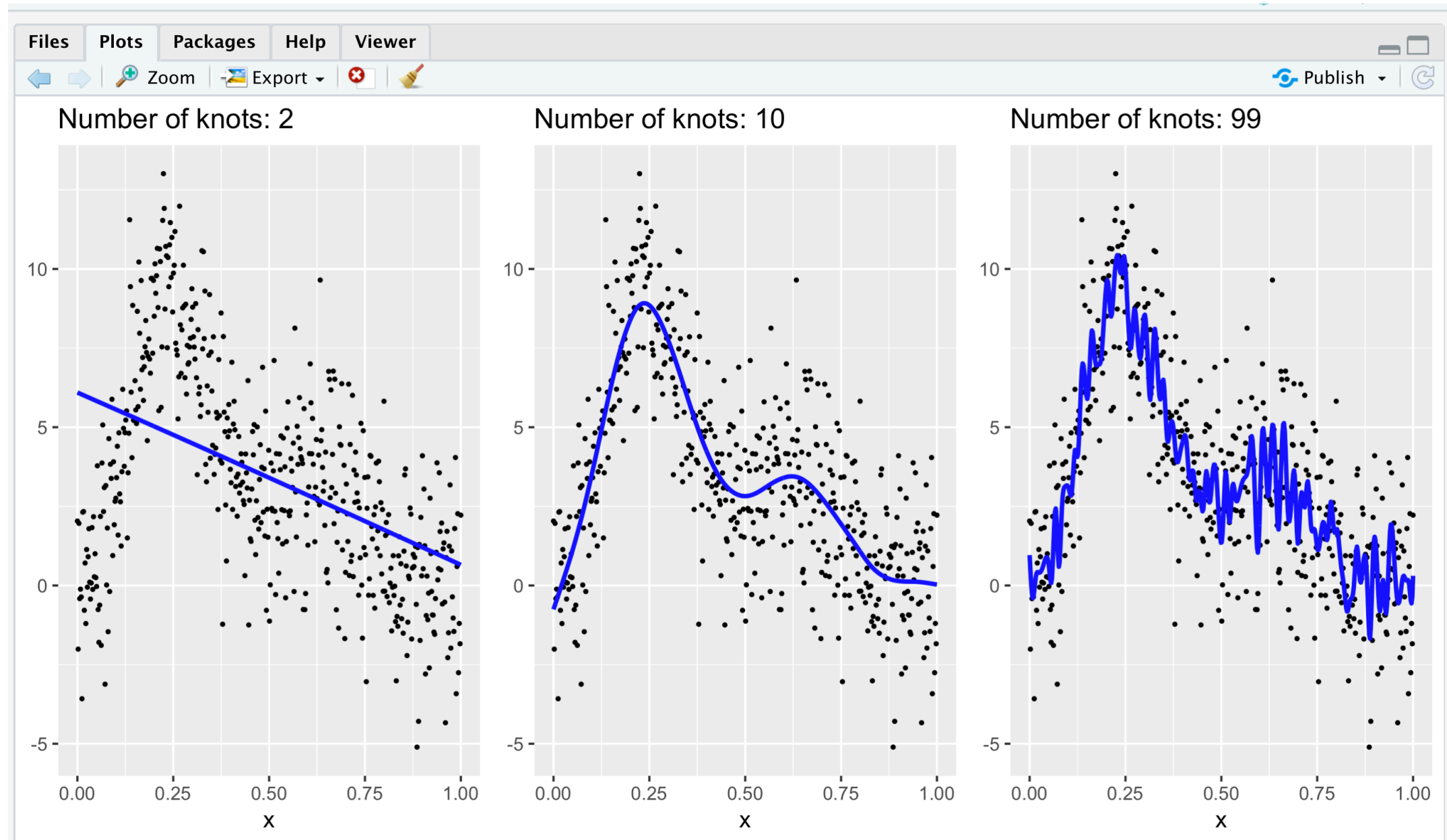
“Just right!”

Overfit!

Machine Learning

- **Goal of Modeling:** Fit models $\hat{f}(x)$ that best approximate the true (unknown) model $f(x)$
- **Goal of Machine Learning:** Fit models that best “predict” the outcome variable

Issue of underfitting vs overfitting?



Underfit!

“Just right!”

Overfit!

Validation Set Approach

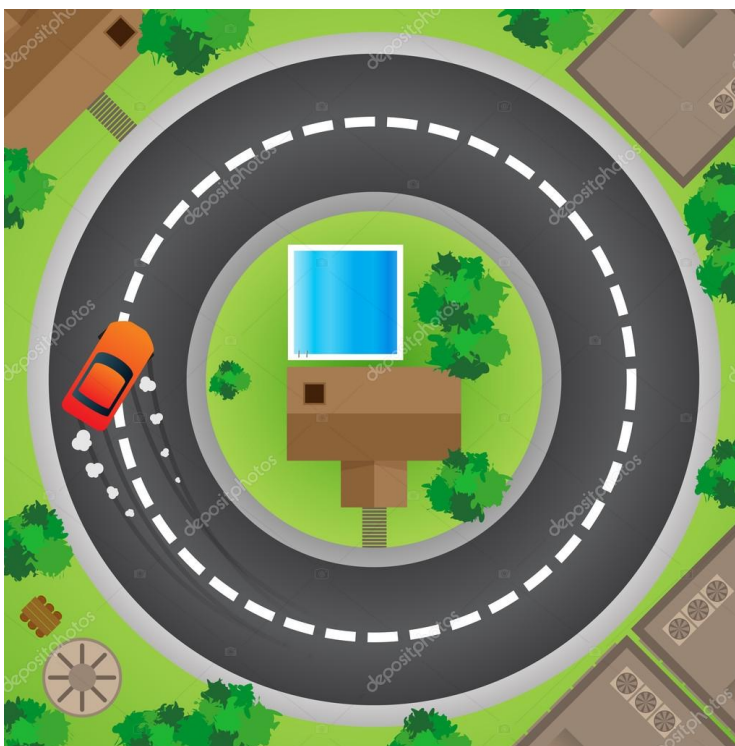


Split your data into:



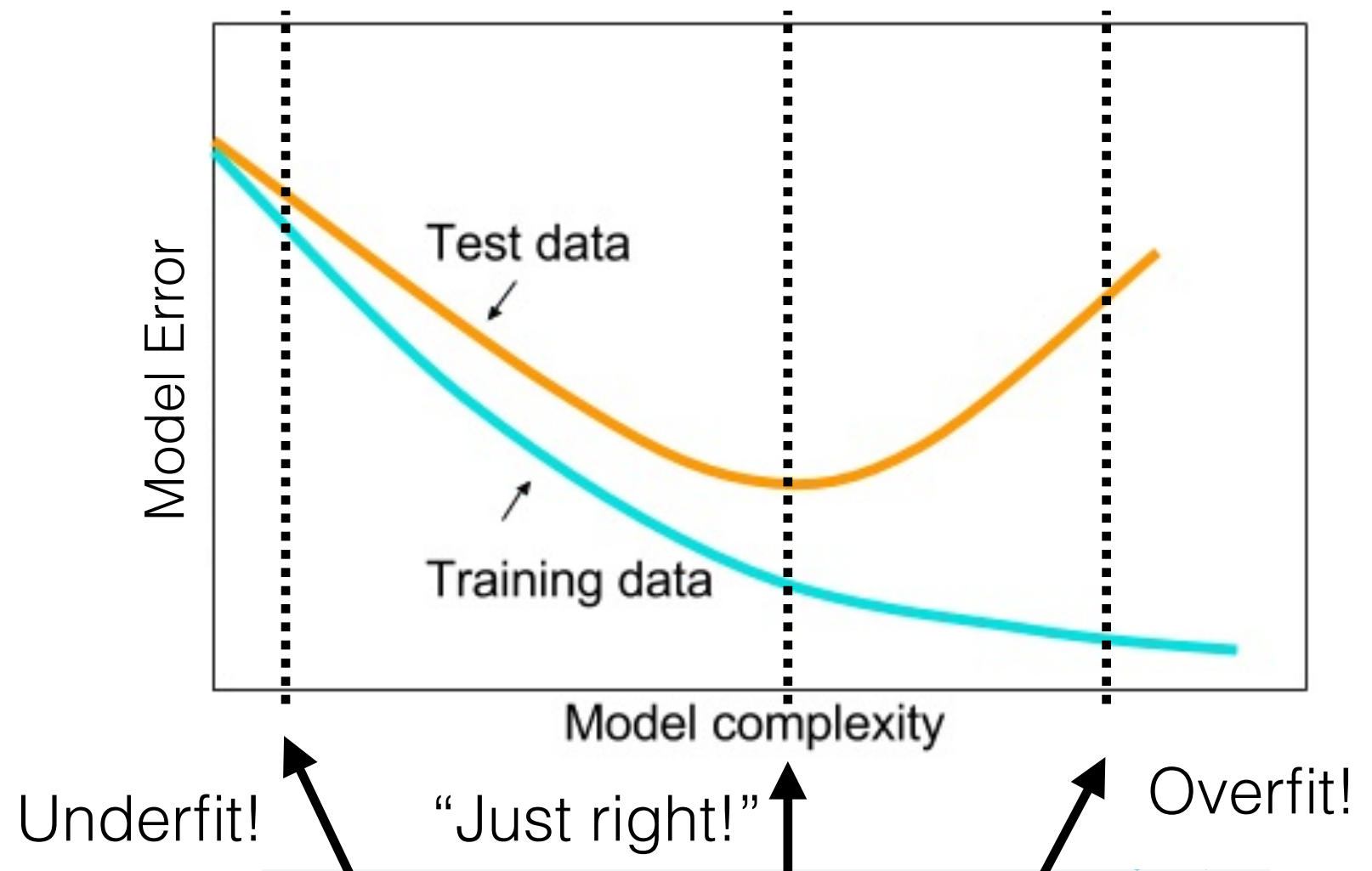
Fit/train model on
training data

Assess model on
independent *test* data

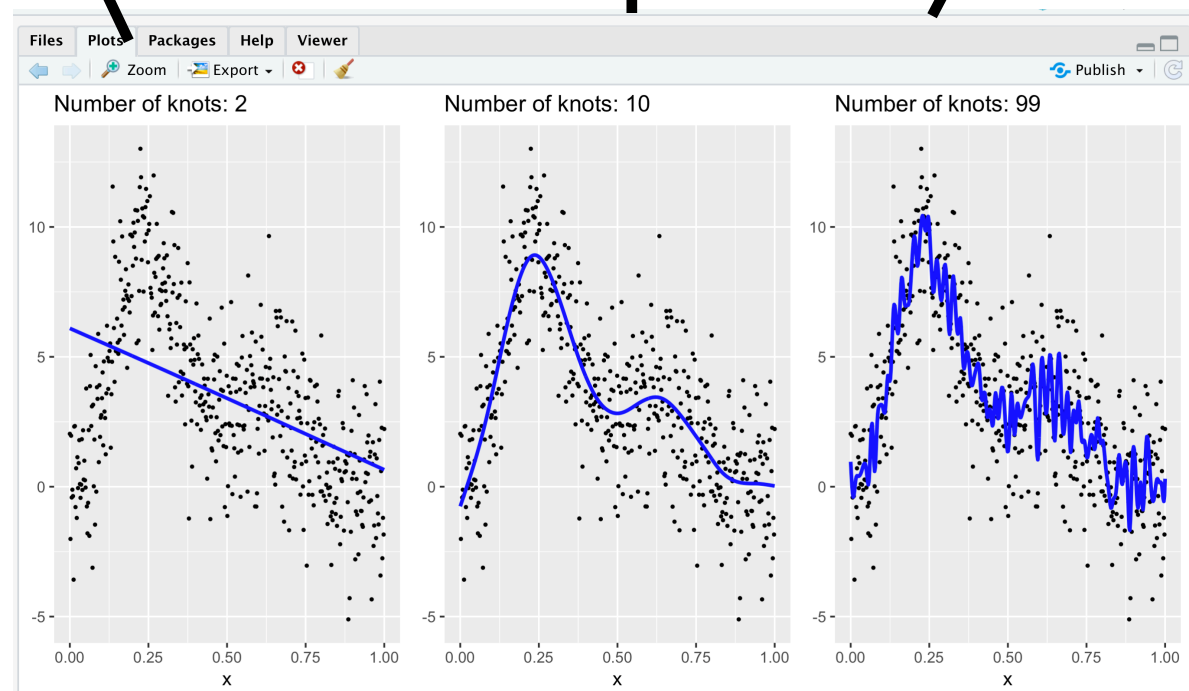


Typical Model Performance

1. Fit/train your model on training data
2. Assess your model error on either:
a) the same training data or
b) independent test data



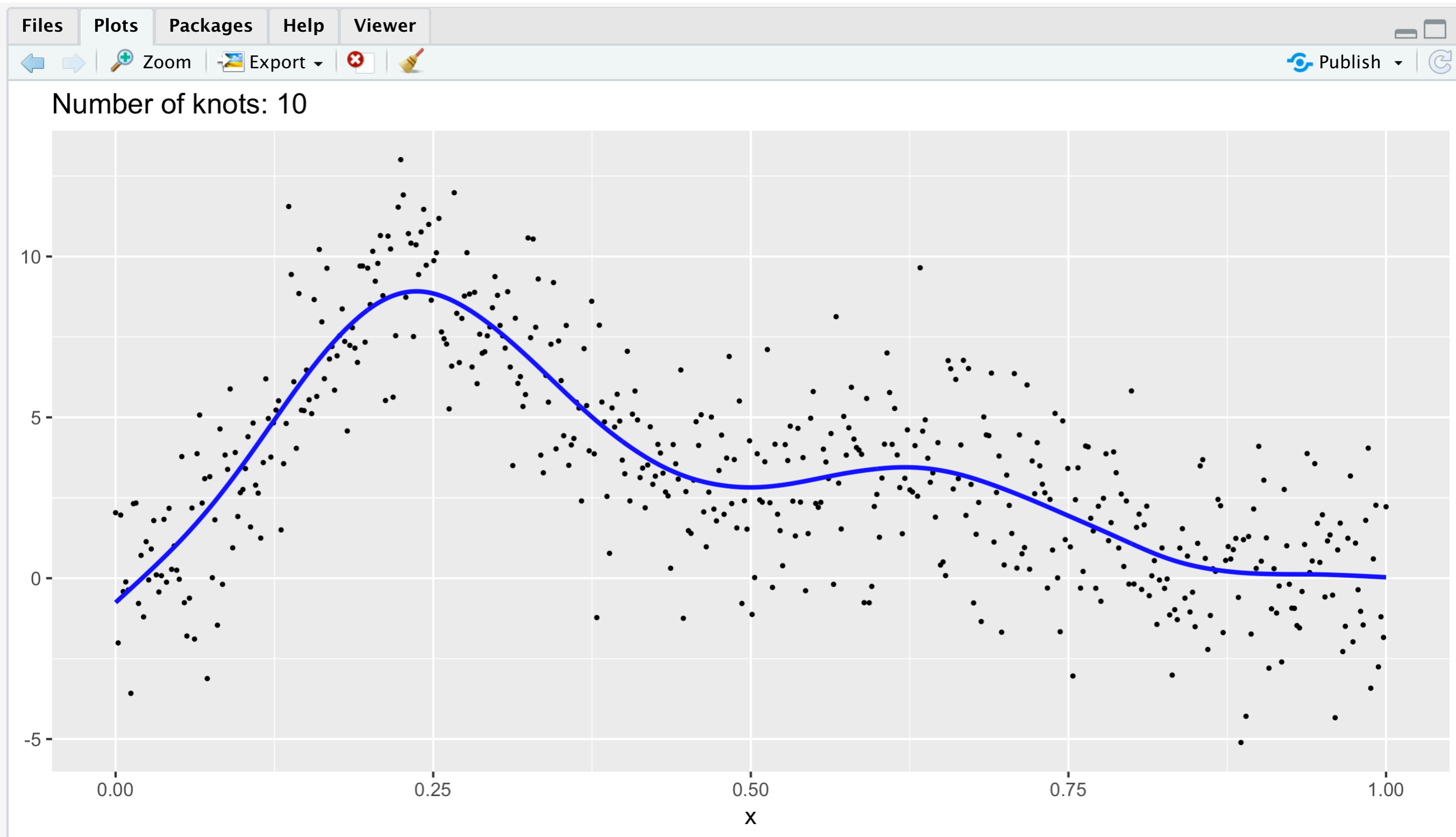
Recall for splines,
the # of knots controls
the **model complexity**



Lastly: A “Wizard of Oz” Reveal...



Our approximated $\hat{f}(x)$...



... was pretty close
to the *true* model:

$$f(x) = 0.2x^{11}(10(1-x))^6 + 10(10x)^3(1-x)^{10}$$

